

KSSANet: KAN-Driven Spatial-Spectral Attention Networks for Hyperspectral Image Super-Resolution

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Abstract—Due to the limitations of physical imaging, acquiring high-resolution hyperspectral images (HR-HSIs) has always been a significant challenge. Single hyperspectral image super-resolution (SHSR) technology aims to generate corresponding HR-HSIs by processing low-resolution hyperspectral images (LR-HSIs). Compared to multi-source data fusion methods, SHSR relies solely on a single low-resolution image and does not require additional auxiliary information or multimodal data, making it more flexible and efficient in data acquisition. Recently, Kolmogorov–Arnold Networks (KAN), which derive from the Kolmogorov–Arnold representation theorem, show great potential in modeling long-range dependencies. In this paper, we further investigate the potential of KAN for hyperspectral image restoration. Specifically, we propose a spatial-spectral attention block (SSAB) module, which includes a KAN-based spatial attention module (KAN-SpaAB) and a KAN-based spectral attention module (KAN-SpeAB), designed for the restoration of spatial and spectral information, respectively. Experimental results demonstrate that KSSANet outperforms existing methods in both quantitative evaluation and image generation quality, achieving state-of-the-art (SOTA) performance. Our code is available at: <https://github.com/Baisongm-Li/KSSANet>.

Index Terms—hyperspectral image, super resolution, Kolmogorov–Arnold Networks (KAN), attention module.

I. INTRODUCTION

Hyperspectral images (HSIs) widely applied in various fields [1]–[7] due to their rich spectral and spatial information. However, the limitations of physical imaging technology make it challenging to directly acquire high resolution hyperspectral images (HR-HSIs). To address this issue, hyperspectral image super-resolution (HSI-SR) techniques have been developed, aiming to generate HR-HSIs from known low-resolution hyperspectral image (LR-HSIs). The main advantage of SHSR is that it does not require additional auxiliary data, and can enhance spatial resolution using only a single hyperspectral image, offering greater independence and flexibility in application. Currently, SHSR methods can be broadly categorized into two types: traditional approaches and deep learning (DL)-based approaches.

Traditional HSI-SR methods typically rely on handcrafted priors, such as low-rank approximation [8]–[12] and sparse coding [13]–[17], to construct mapping functions from LR-HSIs to HR-HSIs. These methods heavily depend on prior

knowledge of the images or scenes, often requiring manually designed features or assumed constraints. However, given the complex and variable nature of HSIs, acquiring accurate prior knowledge is challenging, which limits the performance of traditional approaches. Over the past decade, DL-based methods have developed rapidly, with CNN-based SR networks [18]–[22] achieving significant breakthroughs due to their strong spatial feature modeling capability. However, CNNs face the limitation when reconstructing long-range spectral information in HSIs. To address this issue, Transformer-based networks [23]–[27] have been introduced into the HSI-SR task. The self-attention mechanism of Transformers enables them to capture long-range dependencies and provide robust spectral sequence modeling capability. However, the introduction of Transformer also brings new challenges. First, Transformer-based methods typically require large amounts of training data to achieve stable convergence, but the available HSIs for training are extremely scarce. Second, for a spectral token of length n , the self-attention mechanism has a complexity of $O(n^2)$, which results in an unacceptable computational cost when processing complex high-dimensional HSIs.

Recently, the Kolmogorov–Arnold Network (KAN) [28], [29], a powerful new neural network architecture, has been introduced. Unlike traditional architectures, KANs apply activation functions to the connections between nodes, and these functions can be learned and adapted during training. KANs exhibit strong local plasticity and can avoid catastrophic forgetting by leveraging the locality of splines (a single sample affects only a few nearby spline coefficients).

Specifically, KSSANet achieves its super-resolution capability through multiple cascaded Spatial-Spectral Attention Blocks (SSAB). Each SSAB consists of two key submodules: a KAN-based Spatial Attention Block (KAN-SpaAB) and a KAN-based Spectral Attention Block (KAN-SpeAB). KAN-SpaAB is mainly responsible for reconstructing and optimizing spatial details, while KAN-SpeAB focuses on restoring spectral features. In SSAB, KAN-SpaAB dynamically adjusts the importance scores of spatial pixels, enabling the network to better capture and restore subtle spatial features. Meanwhile, KAN-SpeAB optimizes the importance scores of spectral channels, ensuring that the spectral information of each channel is accurately represented. This joint adjustment mechanism

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allows KSSANet providing precise balancing of spatial and spectral information during reconstruction. Extensive experimental results demonstrate that KSSANet not only excels in subjective visual quality but also significantly surpasses existing SR methods in objective quantitative metrics.

II. PROPOSED METHOD

A. Overall structure

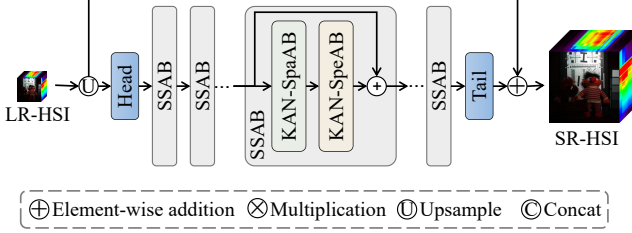


Fig. 1. Overview of the proposed KSSANet.

We denote $\mathbf{I}_{LR} \in \mathbb{R}^{h \times w \times C}$ as the input LR-HSI, $\mathbf{I}_{SR} \in \mathbb{R}^{H \times W \times C}$ as the output HR-HSI, and $\mathbf{I}_{HR} \in \mathbb{R}^{H \times W \times C}$ as the ground truth (GT) for the input image $\mathbf{I}_{LR}$ ($C \gg c$). Our goal is to predict the $\mathbf{I}_{SR}$ from the input $\mathbf{I}_{LR}$ using the proposed HSI-SR network,

$$\mathbf{I}_{SR} = H_{Net}(\mathbf{I}_{LR}), \quad (1)$$

H_{Net} denotes the mapping function of the proposed KSSANet. The overall structure of KSSANet is presented in Fig. 1. KSSANet first uses a *Head* module to project the LR-HSI into latent features. These latent features are then sequentially processed through several cascaded *SSABs*. Finally, a *Tail* module reconstructs the latent features into the shape of the HR-HSI as the predicted image,

$$\mathbf{X}_0 = f_{Head}(\mathbf{I}_{LR} \uparrow), \mathbf{X}_i = f_{SSAB}^i(\mathbf{X}_{i-1}), \mathbf{I}_{SR} = f_{Tail}(\mathbf{X}_i). \quad (2)$$

Let f_{Head} denote the mapping of the *Head* module, which include a 3×3 convolution, $\mathbf{I}_{LR} \uparrow$ represents the bicubic upsampling of LR-HSI. Let f_{SSAB}^i represent the mapping of the i -th *SSAB* module, and f_{Tail} denote the mapping of the *Tail* module, which consists of two 3×3 convolutions, with the final convolution mapping the dimension to the number of spectral bands. $\mathbf{X}_i \in \mathbb{R}^{H \times W \times D}$ represents the latent features output by the i -th *SSAB* module and the number of *SSAB* denoted as L .

B. Kolmogorov-Arnold Networks (KAN)

KAN [28] originates from the Kolmogorov-Arnold representation theorem [30]–[32] and involves a stack of nonlinear, learnable activation functions that reshape the learning process in neural networks. In the KAN layer, each node is fully connected to every node in the subsequent layer. For each edge, an independent, trainable activation function is applied. At each node, only a summation operation is performed over all incoming edges. The learnable activation functions are defined as weighted sums of B-splines, with the B-spline [33], [34] basis functions denoted as B_i , and the fixed residual function chosen as the Sigmoid Linear Unit (SiLU),

$$f_{KAN}(x) = w_1 \cdot \text{SiLU}(x) + w_2 \cdot \sum_{i=0}^{G+k-1} c_i \cdot B_i(x). \quad (3)$$

The weights w_1, w_2 and the basis function coefficients c_i are trainable parameters of spline. The basis function B_i is chosen as a k -th

degree polynomial, with the default value $k = 3$. The grid parameter G determines the degree of B-spline construction with the default value $G = 5$. Although spline-based KANs have ideal mathematical theories, original KAN still struggle with effective HSIs restoration.

C. KAN-SpaAB

We design a KAN-based spatial attention block (KAN-SpaAB) for reconstructing spatial details of HSIs, which is expressed as:

$$\mathbf{X}_{Spa} = f_{Spa}(\mathbf{X}_{i-1}), \quad (4)$$

f_{Spa} represents the mapping of the KAN-SpaAB, and $\mathbf{X}_{Spa}$ represents the output of the KAN-SpaAB. Details as the left part of Fig. 2. To alleviate the limitations of small-kernel convolutions, such as their limited receptive field and inadequate global spatial modeling capability, KAN-SpaAB draws inspiration from the development of large-kernel convolutions [22], [35]. It employs a cascaded combination of decomposed $\text{Conv}5 \times 5$ and $\text{Conv}3 \times 3$ layers to approximate a $\text{Conv}11 \times 11$. The output features from these convolutions are concatenated along the channel dimension, then passed through a depthwise convolution (DWConv) [36], which reduces the channel dimension to 1, resulting in an $H \times W \times 1$ spatial score matrix. Leveraging the strong local plasticity of the KAN network, this spatial score matrix is further refined through two KAN layers using spline transformations. Finally, the spatial matrix is multiplied with the input $\mathbf{X}_{i-1}$ along the spatial dimension, followed by a residual operation. Next, the output of KAN-SpaAB, $\mathbf{X}_{spa}$, is fed into KAN-SpeAB for spectral channel modeling.

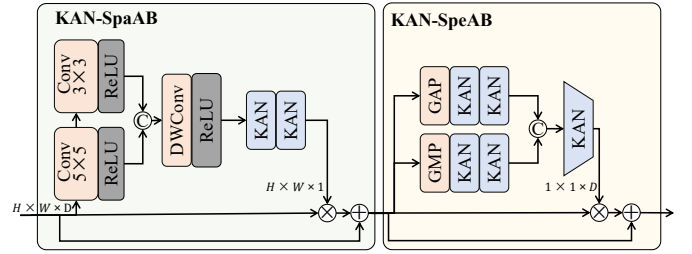


Fig. 2. Structure diagrams of KAN-SpaAB (left) and KAN-SpeAB (right). “Conv3x3” and “Conv5x5” denote 2D convolutions with kernel sizes of 3 and 5, respectively. “ReLU” refers to the ReLU activation function, while “DWConv” represents depthwise separable convolution. “GAP” and “GMP” stand for global average pooling and global max pooling, respectively.

D. KAN-SpeAB

The KAN-based spectral attention block (KAN-SpeAB) is formulated as follows,

$$\mathbf{X}_i = f_{Spe}(\mathbf{X}_{spa}), \quad (5)$$

and the detailed structure is shown on the right side of Fig. 2. Inspired by SENet [37], we utilize globally pooled spatial features as channel excitation features. Specifically, to retain both prominent and global features of the hyperspectral image, we apply global max pooling (GMP) and global average pooling (GAP) to extract two distinct sets of channel excitation features (both size $1 \times 1 \times D$). These excitation features are then refined through two KAN layers using spline adjustments. Afterward, the two features are concatenated along the channel dimension (forming a feature of size $1 \times 1 \times 2D$) and fused via an alignment KAN layer (output feature of size $1 \times 1 \times D$). This produces the final channel importance score matrix, which is multiplied by the input $\mathbf{X}_{spa}$ along the channel dimension, followed by a residual operation. The output of the entire *SSAB* module is denoted as $\mathbf{X}_i$, as followed by a residual operation.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE CHIKUSEI, CAVE AND PAVIA DATASETS. THE BEST VALUES ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST VALUES ARE UNDERLINED.

Methods	Scale	Chikusei				Pavia				CAVE			
		PSNR↑	SAM↓	ERGAS↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	SSIM↑
Bicubic	×2	43.21	1.78	3.59	0.972	34.41	5.28	4.48	0.938	38.06	3.23	4.95	0.966
GDRRN [38]	×2	46.54	1.37	2.58	0.987	37.48	4.77	3.28	0.965	40.97	3.74	4.21	0.973
SSPSR [21]	×2	47.44	1.20	2.28	0.989	37.72	4.65	3.17	0.966	41.38	3.14	3.33	0.975
MCNet [39]	×2	46.78	1.33	2.43	0.987	37.28	4.78	3.31	0.963	41.97	2.66	3.14	0.976
Bi-3DQRNN [40]	×2	45.71	1.43	2.74	0.984	36.90	4.73	3.44	0.962	40.72	2.88	3.60	0.974
DCM-NET [22]	×2	48.02	1.11	2.17	0.990	37.18	5.24	3.37	0.960	41.98	2.70	3.12	0.977
ESSAFormer [26]	×2	<u>48.28</u>	<u>1.10</u>	<u>2.12</u>	<u>0.991</u>	<u>38.78</u>	<u>4.55</u>	<u>2.88</u>	<u>0.971</u>	<u>42.21</u>	<u>2.66</u>	<u>3.04</u>	<u>0.977</u>
KSSANet	×2	48.66	1.02	1.98	0.992	38.92	4.21	2.69	0.982	42.64	2.42	2.92	0.979
Bicubic	×4	37.63	3.40	6.75	0.895	29.67	6.93	7.58	0.815	33.04	4.79	7.84	0.920
GDRRN [38]	×4	39.64	2.63	5.39	0.935	30.84	6.59	6.76	0.867	34.89	4.38	6.75	0.938
SSPSR [21]	×4	40.36	2.35	4.98	0.941	30.64	6.40	6.77	0.861	35.34	4.16	6.50	0.943
MCNet [39]	×4	39.55	2.78	5.36	0.931	30.93	6.68	6.57	0.862	35.58	<u>3.71</u>	6.39	<u>0.947</u>
Bi-3DQRNN [40]	×4	39.89	2.52	5.19	0.937	30.42	7.23	7.03	0.852	35.32	3.92	6.51	0.943
DCM-NET [22]	×4	40.51	2.30	4.85	0.946	30.60	7.26	6.81	0.850	35.50	3.94	6.40	0.944
ESSAFormer [26]	×4	<u>40.76</u>	<u>2.21</u>	<u>4.72</u>	<u>0.948</u>	<u>31.61</u>	<u>6.30</u>	<u>6.10</u>	<u>0.884</u>	<u>35.89</u>	<u>3.83</u>	<u>6.16</u>	<u>0.946</u>
KSSANet	×4	41.02	2.12	4.61	0.952	31.96	5.99	6.01	0.889	35.98	3.69	6.04	0.949
Bicubic	×8	34.50	5.04	9.69	0.806	26.60	8.48	10.77	0.650	29.24	6.60	12.26	0.832
GDRRN [38]	×8	35.22	4.63	9.07	0.835	26.78	9.21	10.57	0.671	30.30	7.15	11.03	0.851
SSPSR [21]	×8	35.82	4.02	8.31	0.853	26.84	10.17	10.47	0.669	31.12	5.51	10.18	0.874
MCNet [39]	×8	35.26	4.61	8.74	0.832	27.03	8.71	10.24	0.680	31.32	5.73	10.02	0.880
Bi-3DQRNN [40]	×8	35.62	4.22	8.49	0.845	26.98	<u>8.47</u>	10.31	0.680	31.17	5.34	10.13	0.879
DCM-NET [22]	×8	35.98	3.93	8.14	0.858	25.65	13.46	12.01	0.548	31.37	5.30	9.93	0.882
ESSAFormer [26]	×8	<u>36.14</u>	<u>3.89</u>	<u>8.11</u>	<u>0.859</u>	<u>27.11</u>	<u>8.91</u>	<u>10.16</u>	<u>0.691</u>	<u>31.43</u>	<u>5.30</u>	<u>9.90</u>	<u>0.884</u>
KSSANet	×8	36.32	3.71	7.96	0.962	27.33	8.40	10.02	0.701	31.51	5.28	9.88	0.888

III. EXPERIMENT

A. Experiment Settings

1) *Datasets, Baselines and Metrics*: Three publicly available hyperspectral datasets are used to validate the effectiveness of the proposed method, including the CAVE [38] natural image dataset and two remote sensing hyperspectral datasets, Chikusei [41] and Pavia Center [42]. The proposed method is compared against six representative deep learning approaches: GDRRN [38], SSPSR [21], MCNet [39], Bi-3DQRNN [40], DCM-NET [22], and ESSAFormer [26]. In all experiments, we employ four metrics to evaluate the performance of all models: PSNR, SAM [43], ERGAS [44], and SSIM [45].

2) *Implementation Details*: We implement KSSANet using the PyTorch framework and uses Adam [46] as the optimizer. The loss function is $L1$ function. The learning rate is set to 1×10^{-4} , with a batch size of 32. The model is trained for 40 epochs. All experiments are conducted on an A800 GPU. In KSSANet, the hidden dimension D is set to 128 and the number of SSAB l is 8. For data preprocessing, we adopt the same approach as used in SSPSR [21]. All efficiency metric results for all models are measured using thop¹.

B. Quantitative and Qualitative Results

The quantitative results for the different methods are demonstrated in Table I. Our proposed KSSANet outperforms existing methods on all metrics across all three scaling factors.

1) *Experiments on the CAVE Dataset*: To validate the effectiveness of KSSANet on natural image datasets, we train our model on the CAVE dataset, randomly selecting 24 images for the training set and 8 images for the test set. The quantitative results are shown in Table I, where KSSANet outperforms existing methods on all metrics. Fig. 3 compares the pseudo-color hyperspectral images generated by various methods and ground truth (GT). While all methods achieve good visual results, KSSANet demonstrates the best spatial detail reconstruction. The mean squared error (MSE) maps in the second

row further show that KSSANet produces the smallest deviation from the GT, resulting in the highest visual quality.

2) *Experiments on the Chikusei and Pavia Dataset*: To validate the effectiveness of KSSANet on remote sensing HSIs, we further test it on Chikusei and Pavia Center datasets. As shown in Table I, KSSANet surpasses all existing methods across all quantitative metrics on these datasets. Additionally, Fig. 4 presents the pseudo-color images and corresponding MSE heatmaps of the high-resolution Chikusei images generated by all methods with a scaling factor of 4. These images demonstrate that KSSANet maintains the best spatial reconstruction capability for remote sensing images, achieving results closest to the GT. In summary, both quantitative and qualitative results indicate that KSSANet outperforms all existing methods.

3) *Efficiency Analysis*: The results as Table II show that the proposed KSSANet not only has smaller model size (i.e., #Params) and computational complexity (i.e., #MACs) compared to ESSAFormer but also achieves the highest PSNR value. These results indicate that KSSANet achieves a better balance between performance and efficiency compared to existing methods.

TABLE II
QUANTITATIVE COMPARISONS OF COMPUTATIONAL PERFORMANCE ON THE CHIKUSEI DATASET FOR ×4 SR.

Methods	#Params(M)	#MACs(G)	PSNR↑
GDRRN [38]	0.44	6.65	39.65
SSPSR [21]	14.88	42.44	40.36
MCNet [39]	2.17	289.63	39.56
Bi-3DQRNN [40]	1.29	120.97	39.89
DCM-NET [22]	12.61	130.9	40.51
ESSAFormer [26]	11.1	48.65	40.76
KSSANet	4.92	20.15	41.02

C. Ablation study

To validate the effectiveness of the proposed spatial and spectral channel attention modules, we conduct an ablation study on these

¹<https://github.com/Lyken17/pytorch-OpCounter>

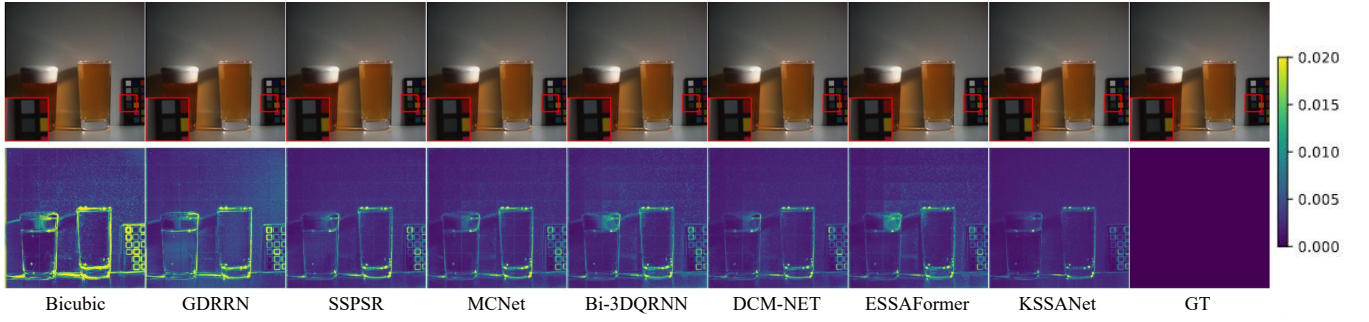


Fig. 3. Visual quality comparison on the CAVE dataset for $\times 4$ SR. The first row presents pseudo-color images (R-29, G-12, B-4), while the second row displays the corresponding MSE heatmaps.

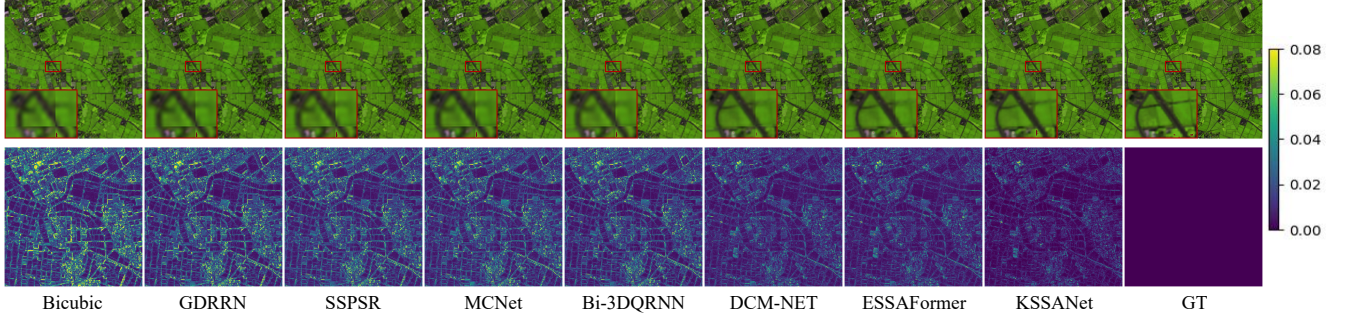


Fig. 4. Visual quality comparison on the Chikusei dataset for $\times 4$ SR. The first row presents pseudo-color images (R-29, G-12, B-4), while the second row displays the corresponding MSE heatmaps.

two modules. Specifically, we replace KAN-SpaAB and KAN-SpeAB with the *Spatial Residual Module* and *Spectral Attention Residual Module* from SSPSR [21], respectively, and retrain and test the model on the Chikusei $\times 4$ dataset. The test results are shown in Table III, although the proposed KAN-SpeAB has larger size and higher computational cost, it delivers significant performance improvements. With the introduction of spline-based KAN, both KAN-SpaAB and KAN-SpeAB demonstrate greater local flexibility, enabling the generation of higher-quality attention matrices, which in turn leads to enhanced performance. To validate the powerful ability

TABLE III

ABLATION COMPARISON ON THE CHIKUSEI DATASET FOR $\times 4$ SR. THE *Spatial Residual Module* AND *Spectral Attention Residual Module* OF SSPSR [21] ARE RESPECTIVELY USED AS ABLATION REPLACEMENTS FOR KAN-SPAAB AND KAN-SPEAB.

KAN-SpaAB	KAN-SpeAB	#Params(M)	#MACs(G)	PSNR $\uparrow$	SSIM $\uparrow$
✓	✓	4.92	20.15	41.02	0.952
✓	✗	0.72	2.89	34.26	0.879
✗	✓	7.29	29.81	39.26	0.928
✗	✗	3.09	12.55	37.89	0.901

of KAN in generating attention matrices, we conduct further ablation experiments on Chikusei $\times 4$ dataset. Specifically, we replace all KAN layers with linear layers to evaluate the advantage of spline-based KAN in terms of local flexibility. After replacing KAN with linear layers, the model size reaches 5.72MB and MACs reach 20.16G, while PSNR and SSIM drop to 38.14dB and 0.907, respectively. Fig. 5 further displays the performance changes during the training process of the two comparison models, showing that the KAN-driven model exhibits more significant improvements in PSNR and reductions in loss value. These experimental results demonstrate the effectiveness of the KAN-driven KSSANet in HSI-SR task.

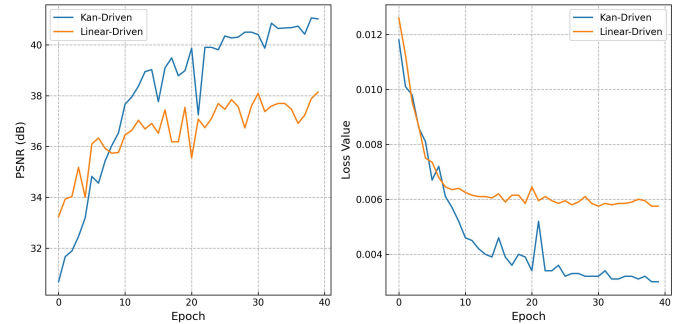


Fig. 5. Performance comparison of KAN-driven and Linear-driven models during training for the $\times 4$ SR on the Chikusei dataset.

IV. CONCLUSION

This paper proposes a KAN-driven single hyperspectral image SR model called KSSANet. Specifically, we leverage the powerful capability of KAN for local adjustments in sequences to generate high-quality attention matrices. We further design a KAN-based spatial attention module (KAN-SpaAB) and a KAN-based spectral attention module (KAN-SpeAB), which are used respectively for image spatial and spectral restoration. Extensive experiments demonstrate that KSSANet not only outperforms existing methods in both quantitative metrics and visual quality.

V. ACKNOWLEDGMENTS

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